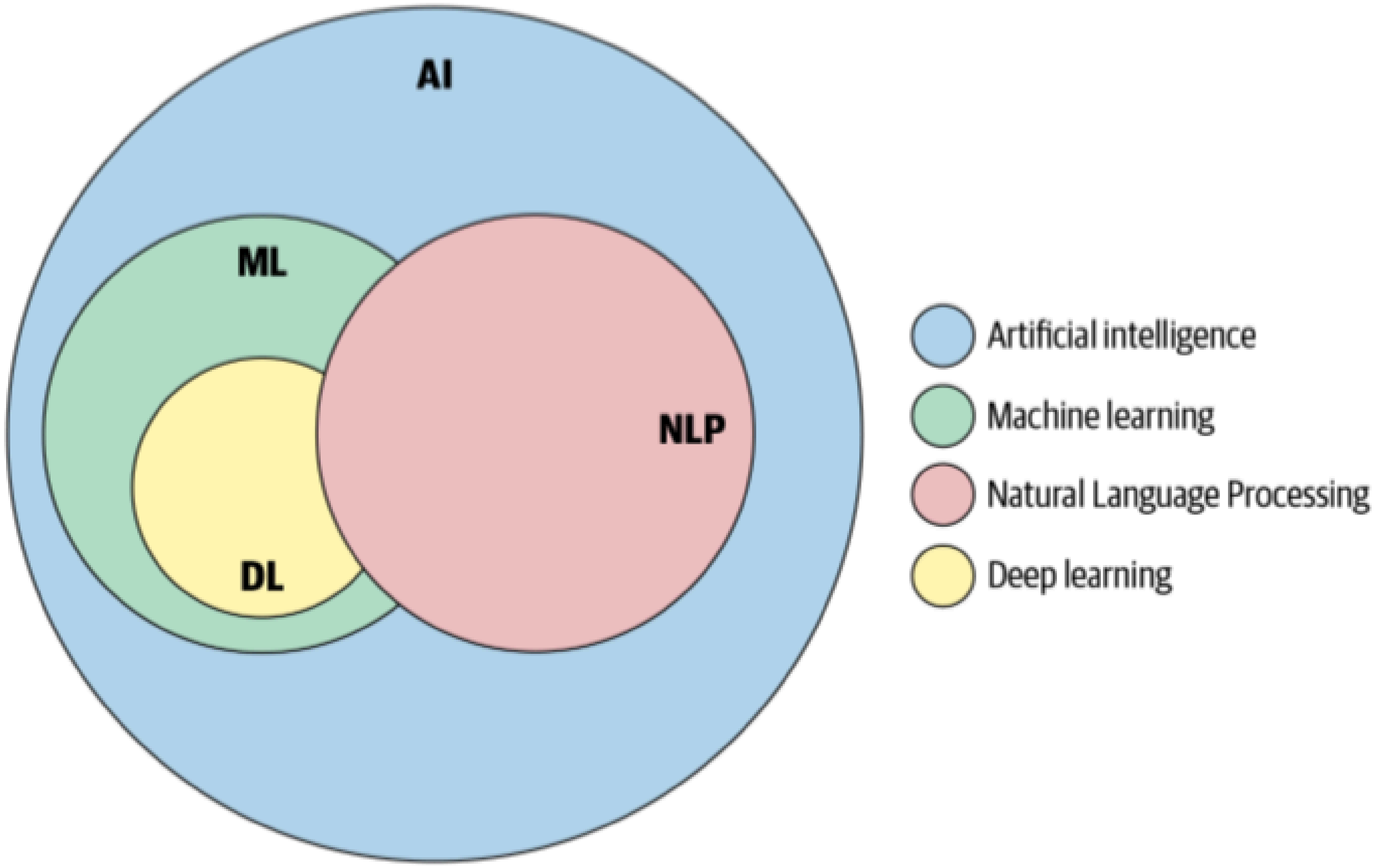
Applied NLP

DATA 641

Lecture 1 — Machine Learning, Deep Learning, and NLP: An Overview

*Figures and notes are from the books Deep Learning with Python, Second Edition by François Chollet, Manning, ISBN 9781617296864 and Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

NLP and its connection to machine learning and artificial intelligence



Artificial intelligence

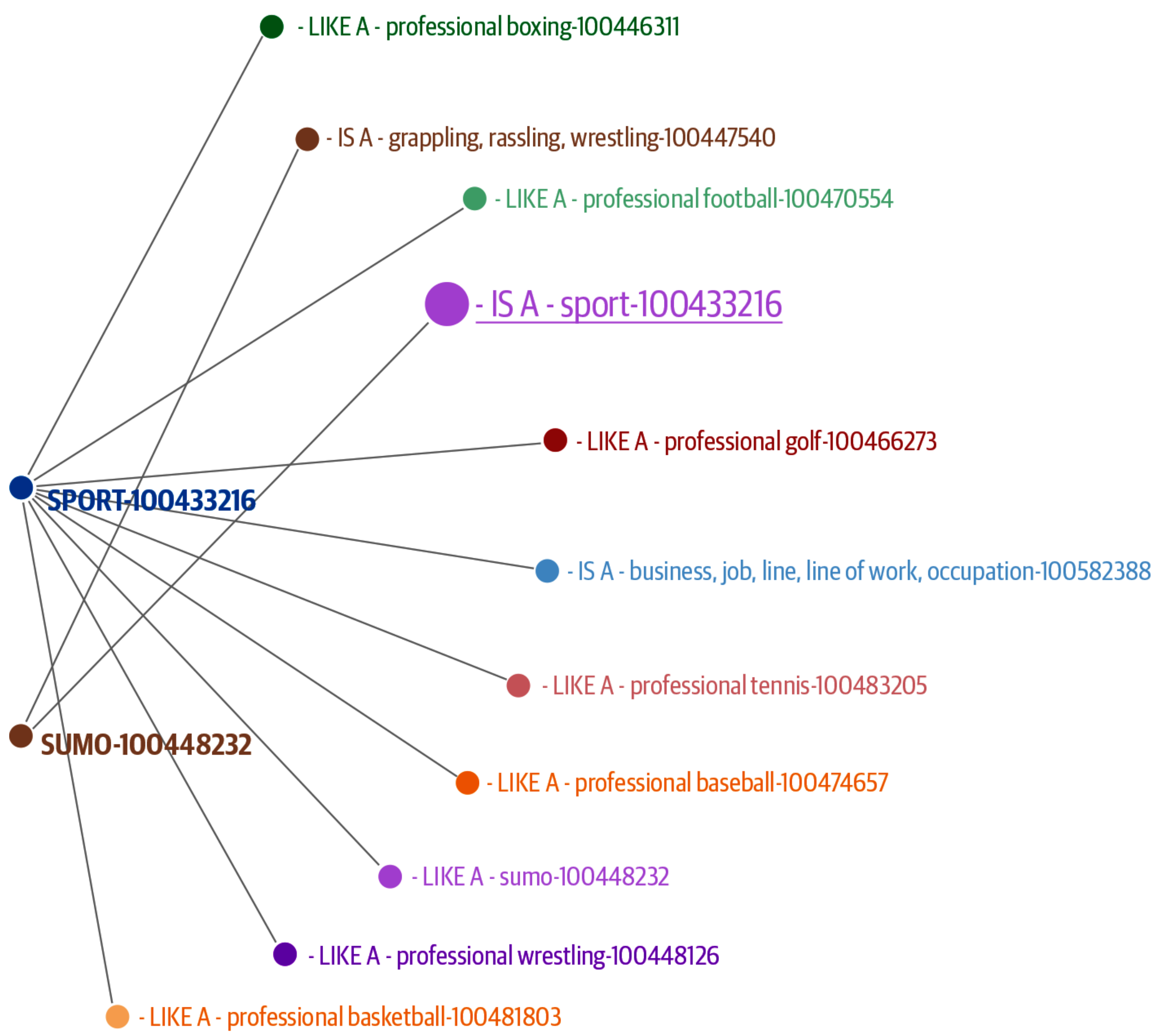
Artificial intelligence (AI) was born in the 1950s, when a handful of pioneers from the nascent field of computer science started asking whether computers could be made to "think"—a question whose ramifications we are still exploring today.

- AI includes many more approaches that may not involve any learning.
    - Early chess programs, for instance, only involved hardcoded rules crafted by programmers, and didn’t qualify as machine learning.
    - For a fairly long time, most experts believed that human-level artificial intelligence could be achieved by having programmers handcraft a sufficiently large set of explicit rules for manipulating knowledge stored in explicit databases.
    - This approach is known as **symbolic AI**. It was the dominant paradigm in AI from the 1950s to the late 1980s, and it reached its peak popularity during the **expert systems** boom of the 1980s.
- Like other early work in AI, early NLP applications were also based on rules and heuristics. In the past few decades, though, NLP application development has been heavily influenced by methods from ML.

Heuristics-Based NLP

Similar to other early AI systems, early attempts at designing NLP systems were based on building rules for the task at hand. This required that the developers had some expertise in the domain to formulate rules that could be incorporated into a program.

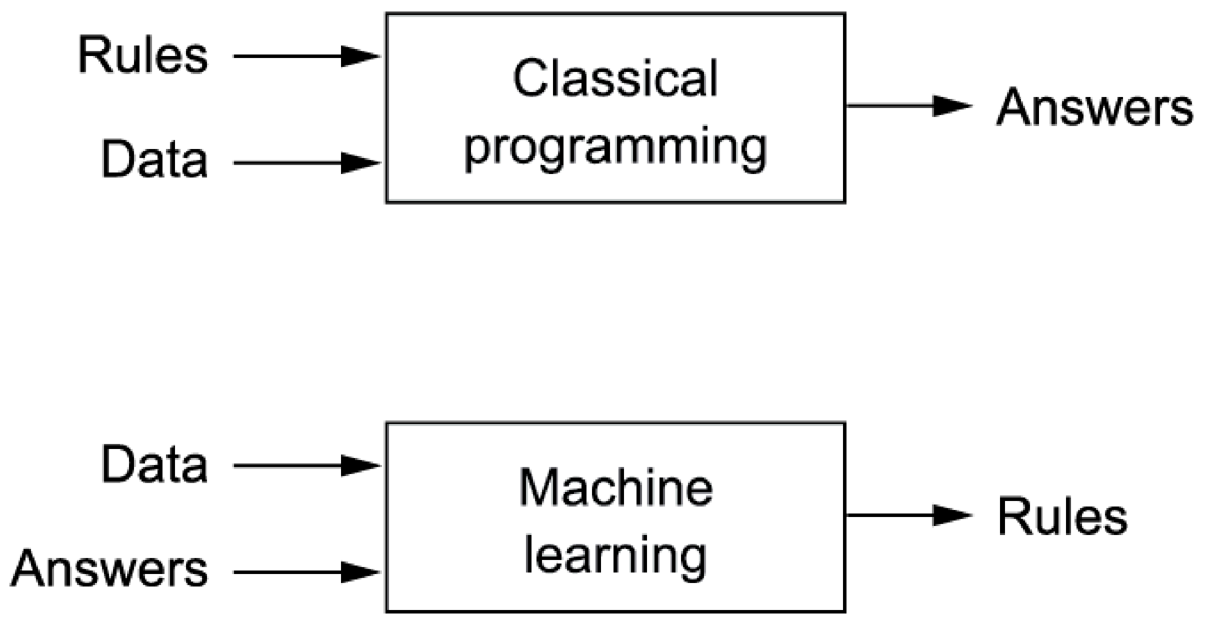
One example is Wordnet which is a database of words and the semantic relationships between them. Some examples of such relationships are synonyms, hyponyms, and meronyms.



Machine learning

The usual way to make a computer do useful work is to have a human programmer write down rules—a computer program—to be followed to turn input data into appropriate answers.

Machine learning turns this around: the machine looks at the input data and the corresponding answers, and figures out what the rules should be. A machine learning system is **trained** rather than **explicitly programmed**.



Learning rules and representations from data

To do machine learning, we need three things:

- **Input data points**. For instance, if the task is speech recognition, these data points could be sound files of people speaking. If the task is image tagging, they could be pictures.
- **Examples of the expected output**. In a speech-recognition task, these could be human-generated transcripts of sound files. In an image task, expected outputs could be tags such as "dog," "cat," and so on.
- **A way to measure whether the algorithm is doing a good job**. This is necessary in order to determine the distance between the algorithm's current output and its expected output. The measurement is used as a feedback signal to adjust the way the algorithm works. This adjustment step is what we call learning.

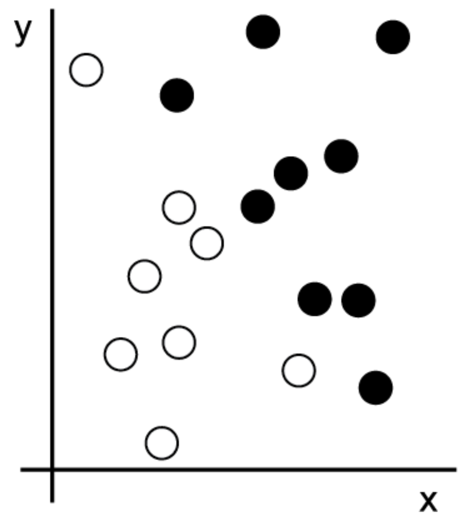
The central problem in machine learning and deep learning is to **meaningfully transform data**: in other words, to learn useful representations of the input data at hand—representations that get us closer to the expected output.

What is a representation?

A color image can be encoded in the RGB format (red-green-blue) or in the HSV format (hue-saturation-value): these are two different representations of the same data.

Machine learning models are all about finding appropriate representations for their input data—transformations of the data that make it more amenable to the task at hand.

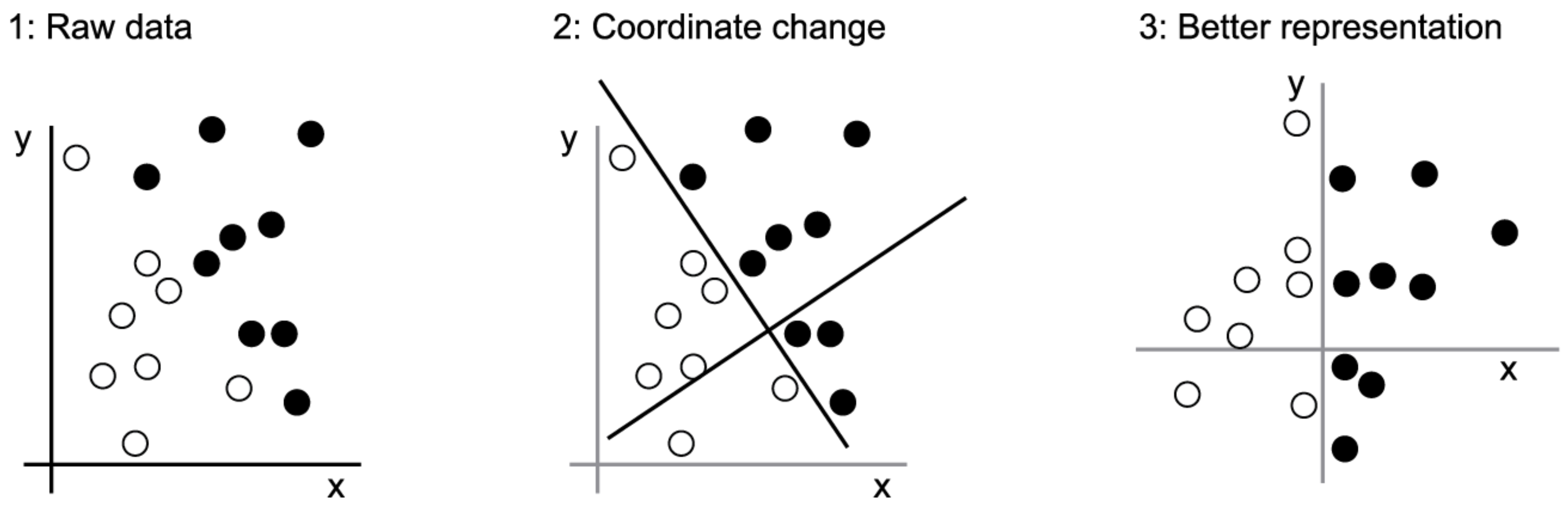
Consider an $x-$axis, a $y-$axis, and some points represented by their coordinates in the (x, y) system.



Let's say we want to develop an algorithm that can take the coordinates (x, y) of a point and output whether that point is likely to be black or to be white. In this case,

- The inputs are the coordinates of our points.
- The expected outputs are the colors of our points.
- A way to measure whether our algorithm is doing a good job could be, for instance, the percentage of points that are being correctly classified.

What we need here is a new representation of our data that cleanly separates the white points from the black points. One transformation we could use, among many other possibilities, would be a coordinate change



With this representation, the black/white classification problem can be expressed as a simple rule: "Black points are such that $x > 0$," or "White points are such that $x < 0$".

Learning, in the context of **machine learning**, describes an automatic search process for data transformations that produce useful representations of some data, guided by some feedback signal—representations that are amenable to simpler rules solving the task at hand.

Some type of transformations can be:

- Coordinate changes
- Taking a histogram of pixels and counting loops (handwriting digits)
- Linear projections
- Translations
- Nonlinear operations (such as "select all points such that $x > 0$")

Machine learning algorithms search through a predefined set of operations, called a hypothesis space. For instance, the space of all possible coordinate changes would be our hypothesis space in the 2D coordinates classification example.

Machine Learning for NLP

Supervised machine learning techniques such as classification and regression methods are heavily used for various NLP tasks.

- An NLP classification task would be to classify news articles into a set of news topics like sports or politics. On the other hand, regression techniques, which give a numeric prediction, can be used to estimate the price of a stock based on processing the social media discussion about that stock.
- Unsupervised clustering algorithms can be used to group together text documents.

Deep learning

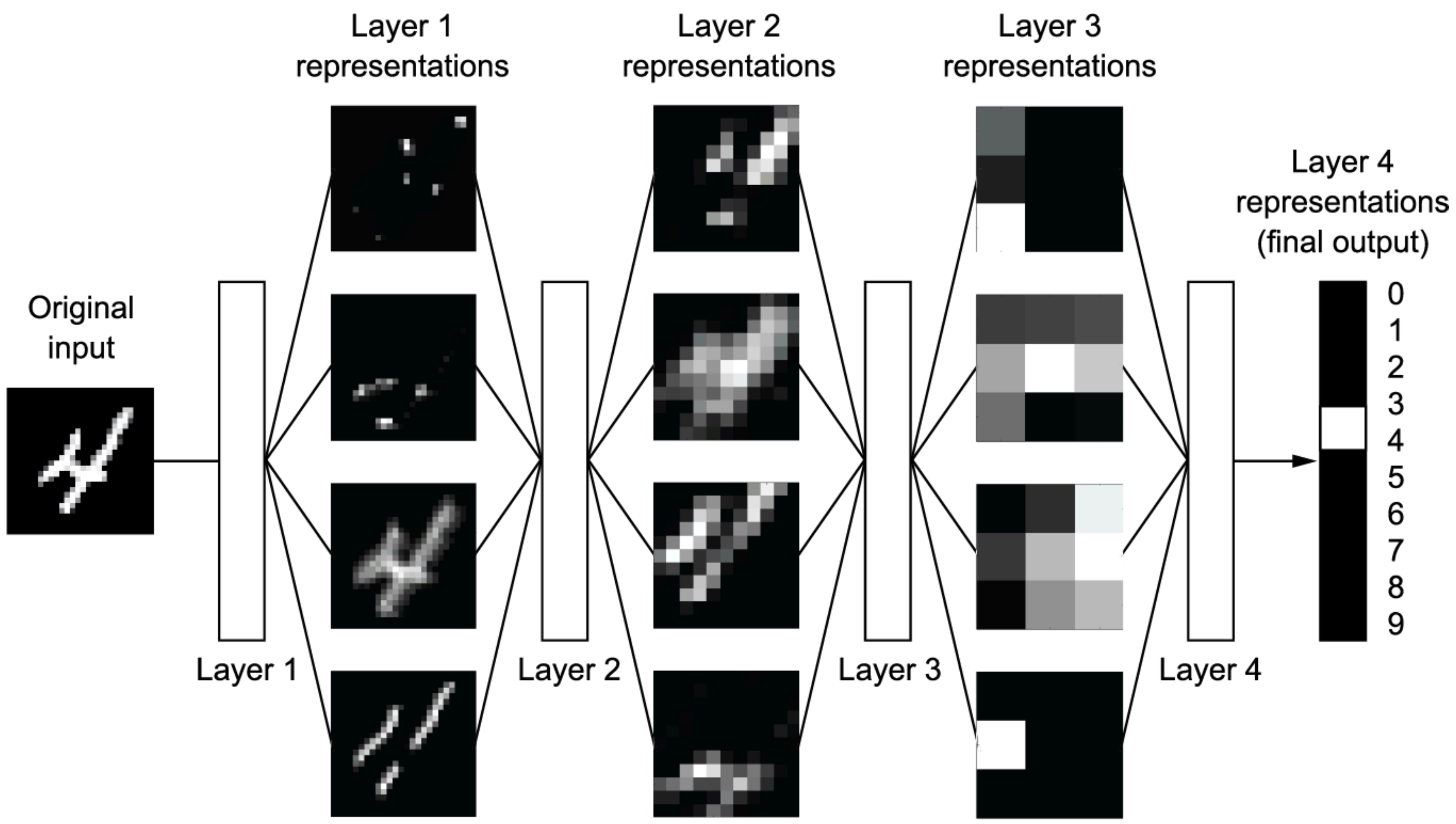
Deep learning is a specific subfield of machine learning: a new take on learning representations from data that puts an emphasis on learning successive layers of increasingly meaningful representations

"deep" in "deep learning" stands for this idea of successive layers of representations. How many layers contribute to a model of the data is called the **depth** of the model.

In deep learning, these layered representations are learned via models called **neural networks**, structured in literal layers stacked on top of each other.

The term **neural network** refers to neurobiology, but although some of the central concepts in deep learning were developed in part by drawing inspiration from our understanding of the brain (in particular, the visual cortex), **deep learning models are not models of the brain**.

What do the representations learned by a deep learning algorithm look like?



Think of a deep network as a multistage information-distillation process, where information goes through successive filters and comes out increasingly purified (that is, useful with regard to some task).

Deep Learning for NLP

In the last few years, we have seen a huge surge in using neural networks to deal with complex, unstructured data. Language is inherently complex and unstructured. Therefore, we need models with better representation and learning capability to understand and solve language tasks. Some popular deep neural network architectures that have become the status quo in NLP include recurrent neural networks, convolutional neural networks, transformers, and autoencoders.